

Mirage AC IR Protocol Documentation v2

EICON121-0 Indoor Unit / KKG27A-C11 Remote

Overview

This document summarizes the reverse engineering of the EICON121-0 mini-split IR protocol using ESPHome and an ESP32.

Features Decoded

Power, Temperature, Modes, Fan Speeds, Turbo, Quiet, Sleep, Health, Clean, Lock, H-Sweep, V-Sweep, Display Toggle, Celsius/Fahrenheit, ON Timer, OFF Timer.

Protocol Structure

Packet Length: 14 Bytes

Byte 0 - Header (0x56)

Byte 1 - Temperature / Setpoint

Byte 2 - Unknown

Byte 3 - Quiet

Byte 4 - Mode + Fan

Byte 5 - Swing / Display / Power State

Byte 6 - Health / Sleep / Heat / Lock

Byte 7 - Clean

Byte 8 - ON Timer Hours

Byte 9 - Temperature Units

Byte 10 - OFF Timer Hours

Byte 11 - Unknown

Byte 12 - Unknown

Byte 13 - Unknown

Temperature Encoding

Byte1 = Temperature + 0x5C

20°C = 0x70

21°C = 0x71

Byte Definitions

Byte 3: 0x01 = Quiet

Byte 4 Modes: 0x10 Heat, 0x20 Cool, 0x30 Dry, 0x40 Auto, 0x50 Fan
Fan Offsets: +0 Auto, +1 High, +2 Low, +3 Medium

Byte 5: 0x01 H-Sweep, 0x02 V-Sweep, 0x03 H+V Sweep, 0x04 Display Toggle, 0xC0 Power Off

Byte 6: 0x02 Health, 0x08 Sleep, 0x40 Heat Flag, 0x80 Lock

Byte 7: 0x40 Clean

Byte 8: ON Timer Hours (0x03=3h, 0x05=5h)

Byte 9: 0x00 Celsius, 0xC0 Fahrenheit

Byte 10: OFF Timer Hours (0x01=1h, 0x05=5h)

Example Packets

Cool 20°C Auto Fan:

56:70:00:00:20:00:00:00:00:00:00:00:00

Power Off:

56:70:00:00:20:C0:00:00:00:00:00:00:00

Turbo:

56:70:00:00:20:00:00:00:80:00:00:00:00

Clean:

56:70:00:00:20:00:00:40:00:00:00:00:00

Remaining Unknown Fields

Byte 2, Byte 11, Byte 12, Byte 13

Protocol Discoveries Timeline

Phase 1: Temperature, Modes, Fan Speeds

Phase 2: Turbo, Quiet, Sleep, Health, Clean, Swing

Phase 3: Celsius/Fahrenheit, Lock, ON Timer, OFF Timer

Phase 4: Investigation of remaining unknown bytes